Tochukwu D. Eze

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RESEARCH INTERESTS

Human-centered AI; multimodal learning analytics; LLM-based feedback and agents; computer-supported collaborative learning; privacy-preserving multimodal sensing; affective computing.

EDUCATION

Northwestern University, Ph.D. in Computer Science March, 2027
Advisor: Prof. Marcelo Worsley. Dissertation: Evidence-Grounded and Privacy-Aware AI for Modeling and Supporting Multi-Party Collaboration

Northwestern University, M.S. in Computer Science June 2023
GPA: 3.963/4.0

Nnamdi Azikiwe University, M.Sc. in Computer Science 2015 – 2018
Distinction (4.71/5.0). Thesis: TPL, a multi-target programming language compiling to several high-level languages

Nnamdi Azikiwe University, B.Sc. in Computer Science 2008 – 2012
First Class Honours (4.69/5.0)

PUBLICATIONS

Journal Articles

1. **Eze, T. D.**, Zurbuchen, L., Anderson, K., & Worsley, M. (2026). A collaboration literacy analytics framework (CLAF): Investigating students' perspectives on collaboration quality indicators and feedback utilization. *International Journal of Computer-Supported Collaborative Learning*. https://doi.org/10.1007/s11412-026-09475-w
2. **Eze, T. D.**, Anderson, K., & Worsley, M. (2026). Evaluating AI-based visual privacy through cartoonification: Impact on expressive participation and consent for data retention. *Computers and Education Open, 11*, 100393. https://doi.org/10.1016/j.caeo.2026.100393

Peer-Reviewed Conference Papers

3. Lin, C., **Eze, T. D.**, Xie, D., Shin, B., Worsley, M., (2026). Unmasking suppressed emotion: A multi-agent approach to affective dissonance. In *Proceedings of the 28th ACM International Conference on Multimodal Interaction (ICMI 2026)*. ACM.
4. Xie, D., Anderson, K., **Eze, T. D.**, Lin, C., Shin, B., & Worsley, M. (2026). CLARA: An AI-augmented analytics dashboard for collaboration literacy. In *Proceedings of the International Conference on Artificial Intelligence in Education (AIED 2026)* (pp. 339–353). Springer.

Under Review

5. **Eze, T. D.**, Xie D., Anderson K., Shin B., Lin C., Puthipiroj P., Worsley M. From analytics to regulation: Designing for collaboration literacy with LLM-enabled dialogic feedback. Submitted to *ACM CHI Conference on Human Factors in Computing Systems (CHI 2027)*.
6. **Eze, T. D.**, Ezema, K., De Meran Meguimtsop, A., & Adejoro, C. Understanding privacy in context: A systematic review of risks, protections, and human-centered challenges in multimodal learning analytics. Submitted to *Journal of Learning Analytics*.

Posters and Presentations

7. **Eze, T. D.** (2025, April 4). AI-based visual privacy and student engagement in sensitive discussion on social justice issues [Poster]. Annual Yale Bouchet Conference on Graduate Education, New Haven, CT.
8. Worsley, M., Anderson, K., **Eze, T. D.**, & Mirsaidova, A. (2023, March). Student perceptions of collaboration analytics [Poster]. Collaborative Analytics Workshop, Learning Analytics and Knowledge Conference (LAK 2023), Arlington, TX.

RESEARCH EXPERIENCE

Graduate Research Assistant, tiilt Lab, Northwestern University Sept 2022 – Present
Advisor: Prof. Marcelo Worsley

- Led an eight-week randomized classroom study (100 students, 175 team sessions) of an evidence-grounded LLM dialogic feedback system built on Group Communication Analysis indicators; dialogic feedback increased movement into interactionally integrated collaboration profiles (79% vs. 43–52%) (under review, CHI 2027).

- Developed the Collaboration Literacy Analytics Framework (CLAF) from a 12-week study of 290 students using a collaboration analytics tool (IJCSCL 2026).
- Built a real-time, expression-preserving cartoonification layer (Dlib, VToonify) for the BLINC video discussion platform and showed it increased expressive participation in sensitive-topic discussions (Computers and Education Open 2026).
- Integrated computer vision gaze tracking into a collaborative learning analytics tool to study sustained and shared attention.
- Contributed to CLARA, an agentic LLM analytics dashboard (AIED 2026), and a multi-agent LLM framework for detecting masked emotion (ICMI 2026); led a systematic review of privacy in multimodal learning analytics.

Graduate Researcher, Racket Lab, Northwestern University Jan 2021-Aug 2022

- Modeled the JavaScript runtime to show that IRIS protects popular libraries from security exploits arising from loopholes in the CommonJS module system.
- Investigated an error-detection tool for ReactJS to catch bugs caused by misuse of React hooks.

Graduate Research Assistant, Nnamdi Azikiwe University Apr 2016 – Nov 2018

- Designed and implemented TPL, a Java-based programming language that compiles to several high-level languages, enabling a single language to target multiple domains.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Northwestern University Jan 2022 – Present

- *CS 214: Data Structures and Algorithms* (5 quarters): led lab sessions, facilitated discussions, graded, and tutored students one-on-one.
- *CS 330: Human-Computer Interaction* (1 quarter): evaluated project work and facilitated user experience design workshops.
- *CS 329: HCI Studio* (1 quarter): supported studio-based interactive system design and prototyping.
- *CS 348: Introduction to Artificial Intelligence* (1 quarter): supported instruction, project guidance, and grading in machine learning, search, and AI ethics.

Assistant Lecturer, Nnamdi Azikiwe University Apr 2015 – Nov 2020

- *CSC 101 & 102: Introduction to Computer Science and Programming I & II* (instructor): taught first-year programming, supervised group projects, and evaluated final presentations.
- *CSC 231: Automata Theory and Computation* (instructor): taught computational theory and led a project in which students designed an interpreter that detects syntax errors.

AWARDS AND HONORS

Recognized among the 100 Most Impactful Educational Personalities in Africa, Global Skills Hub (UK) Jul 2025
Fellowship in Leadership for Graduate Students and Postdocs, Northwestern University May 2025
Graduate Assistant of the Month, Residential Academic Engagement, Northwestern University May 2025
Edward Alexander Bouchet Graduate Honor Society, Northwestern University Chapter Apr 2025
Northwestern University Graduate Fellowship Jan 2021
Dean's Award for Best Student (CGPA 4.0+), Faculty of Physical Sciences, Nnamdi Azikiwe University 2010, 2011

LEADERSHIP AND SERVICE

Assistant Chair, Chapin Residential College, Northwestern University Sep 2023 – Present

- Develop academic and cultural programming with the Faculty Chair, advise the student executive board, and manage the college budget.

Co-President, African Graduate Student Association, Northwestern University May 2024 – Present

- Lead a 7-member board serving 80+ active members on a $1,500 annual budget.

Graduate Leader, English Language Proficiency Program, Northwestern University Sep 2024 – Present

- Organize regular events that give non-native English-speaking international students opportunities to practice English.

Peer Mentor, English Language Proficiency Program, Northwestern University Jun 2024 – Sep 2024

- Mentored newly admitted international students and ran seminars on housing, transportation, and daily life in Evanston and Chicago.

Co-Professional Development Chair, McCormick Graduate Leadership Council Sep 2021 – Sep 2022

- Planned social, academic, and professional programming with a 12-member council.

University ICT Officer; Faculty ICT Officer; Chair, Faculty ICT Committee, Nnamdi Azikiwe University 2015 – 2018

- Coordinated university-wide ICT operations with the Deputy Vice-Chancellor (Academic), supervised departmental ICT officers, and ran training for staff and students on university software systems.

PROFESSIONAL EXPERIENCE

Software Engineer, Nnamdi Azikiwe University Apr 2015 – Nov 2020

- Built and maintained NAUPortal, the university's student records, result verification, biometric, fee payment, computer-based examination, and mobile app system.

Software Consultant, Chukwuemeka Odumegwu Ojukwu University Sep 2014 – Sep 2020

- Developed the SIWES internship placement portal matching students to companies, and trained 2,000+ students annually to use it.

Software Developer, Oliswarren Associates Sep 2012 – Sep 2014

- Built the Anambra State Motor Vehicle Administration System and integrated e-payment verification that increased revenue by over 50% and reduced fraud.

TECHNICAL SKILLS

Research methods: randomized field studies; mixed-effects modeling (R: lme4, mclogit); clustering; thematic analysis; systematic reviews (PRISMA)
AI and machine learning: LLM prompting and agentic pipelines; NLP; Sentence-BERT; PyTorch; TensorFlow; scikit-learn
Multimodal sensing: computer vision (Dlib, Py-Feat, VToonify); speech-to-text and speaker attribution
Programming: Python, R, Java, C++, JavaScript, TypeScript, PHP
Web and systems: React, Angular, HTML/CSS; MySQL, PostgreSQL, SQLite; Apache, Nginx, Tomcat; Linux; Git

OTHER CONFERENCE PARTICIPATION

- Computer Professionals Registration Council of Nigeria (CPN) National Conferences, Abuja, Nigeria (2017, 2018).
- Digital Bridge Institute ADAPTI Workshop on SPSS, Nnamdi Azikiwe University (2017).
- WACREN E-Research Hackfest on Open Science and E-infrastructure, Lagos State University (2016).

PROFESSIONAL MEMBERSHIPS

Association for Computing Machinery (ACM); Society for Learning Analytics Research (SoLAR); Applied AI Association (AAIA)